

Advanced Microeconomics

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Problem Set 2: Theory recap, Applications of Decision Modeling

1. Theory

- a) Give a brief definition of rational choice.
- b) Summarize the requirements to model consumer decisions.
- c) Define the Marginal Rate of Substitution (MRS).

2. Application

- a) Calculate all partial derivatives and the total derivatives for the following functions:

$$1) f(x, y) = x^3 y^3$$

$$2) f(x, y) = \frac{1}{1 + x^4 + y^2}$$

$$3) f(x, y) = e^{x^2 y^2}$$

- b) Assume the utility function $U_1 = 4xy^2$.

- 1) Calculate the MRS.
- 2) Perform a monotone transformation of U_1 and recalculate the MRS.
- 3) Briefly interpret your results.

c) Given is the following utility function $U(x, y) = 4x^2y^2$ and the budget constraint $100 = 8x + 3y$ of a household.

- 1) Draw the budget constraint and briefly describe what it reflects.
- 2) What are the optimum quantities of x and y consumed by a household with the given utility function and budget constraint?
- 3) What is the optimum utility of the household?
- 4) Assume that the price of good x is reduced by 3 monetary units. Calculate the utility of the household in the new optimum.
- 5) How can the change in the level of utility be interpreted?